SSFam: Scribble Supervised Salient Object Detection Family

Zhengyi Liu, Sheng Deng, Xinrui Wang, Linbo Wang, Xianyong Fang, *Member, IEEE*, and Bin Tang

***Abstract*—Scribble supervised salient object detection (SSSOD) constructs segmentation ability of attractive objects from surroundings under the supervision of sparse scribble labels. For the better segmentation, depth and thermal infrared modalities serve as the supplement to RGB images in the complex scenes. Existing methods specifically design various feature extraction and multi-modal fusion strategies for RGB, RGB-Depth, RGB-Thermal, and Visual-Depth-Thermal image input respectively, leading to similar model flood. As the recently proposed Segment Anything Model (SAM) possesses extraordinary segmentation and prompt interactive capability, we propose an SSSOD family based on SAM, named *SSFam*, for the combination input with different modalities. Firstly, different modal-aware modulators are designed to attain modal-specific knowledge which cooperates with modal-agnostic information extracted from the frozen SAM encoder for the better feature ensemble. Secondly, a siamese decoder is tailored to bridge the gap between the training with scribble prompt and the testing with no prompt for the stronger decoding ability. Our model demonstrates the remarkable performance among combinations of different modalities and refreshes the highest level of scribble supervised methods and comes close to the ones of fully supervised methods.**

***Index Terms*—Salient object detection, scribble supervision, multi-modal, segment anything model, unified model.**

I. Introduction

SALIENT object detection (SOD) focuses on the identification and segmentation of most visually prominent objects in RGB images, which benefits the other computer vision tasks, e.g., semantic segmentation [1], [2], [3], object tracking [4], [5], person and vehicle re-identification [6], [7], [8], and audio-visual quality assessment [9].

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[Online]. Available: https://github.com/liuzywen/SSFam.

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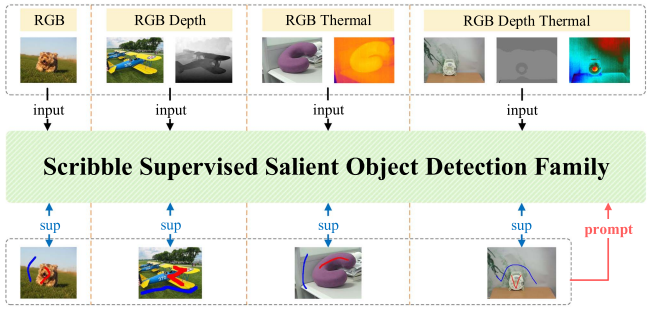


Fig. 1. Scribble supervised salient object detection family for unimodal, bimodal, and trimodal images.

The prevalent fully supervised methods design and train deep learning model with high-cost pixel-level annotations, while the Scribble Supervised Salient Object Detection (SSSOD) methods [10] use the pair of input image and sparse scribble labels indicating foreground and background to train the model and use input image for testing, showing the potential due to less human intervention and moderate supervision.

Existing SSSOD methods [11], [12], [13] borrow pretrained CNN and Transformer encoders which are originally designed for classification task, and then cooperate with progressively upsampling decoders. The performance is restrained due to the gap between classification task and segmentation one.

The recent proposed large model Segment Anything Model (SAM) [14] pretrained for segmentation tasks is naturally a better choice to solve SSSOD task. Some pioneering researches [15], [16], [17], [18], [19], [20], [21], [22], [23] have fine-tuned SAM to specific unimodal tasks via parameter-efficient transfer learning techniques [24].

However, in real-world scenes, RGB cameras, depth and thermal infrared sensors are combinational equipped to improve visual perception ability. Accordingly, bimodal SOD, such as RGB-D [25], [26], [27], [28] and RGB-T [29], and trimodal visible-depth-thermal (V-D-T) SOD [30] are tailored to suit the interaction requirements of different modalities, which increase modeling overhead.

Facing the input challenge with combinations of different modalities, we propose an SSSOD family for unimodal, bimodal, and trimodal input, named *SSFam* (Fig. 1). Some modal-aware modulators are attached to SAM and responsible for obtaining modal-specific features, which will be aggregated into

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the modal-agnostic features from the frozen SAM encoder by element-wise scaled sum for a better feature ensemble.

Besides, the large model SAM supports point, box, and mask prompt to generate the class-agnostic mask by indicating object position. VRP-SAM [31] designs a visual reference prompt to extend the visual reference segmentation capabilities of SAM. The scribbles in the SSSOD task supervise the training of the model in the training stage, and meanwhile they can be regarded as prompt of SAM. Concretely, some points are randomly extracted from the scribbles and then fed into the prompt encoder of SAM, named scribble prompt, which extending the weakly supervised learning capabilities of SAM. However, there is no prompt in the testing stage. To eliminate the gap, a siamese decoder is proposed to transfer the parameters with prompts in the training to the ones with no prompt in the testing.

In summary, the main contributions are summarized as follows:

- To the best of our knowledge, we are the first to use the large model SAM to solve scribble supervised multi-modal salient object detection tasks involved RGB, RGB-D, RGB-T, V-D-T combinations of multiple modalities.
- To reduce the number of trainable parameters, modal-aware modulators are designed to extract modal-specific features besides modal-agnostic features from the frozen SAM encoder.
- To eliminate the gap between scribble prompt training and no prompt testing, a siamese decoder is proposed to transfer the parameters with prompts in the training to the ones with no prompt in the testing.
- Experiments show our approach significantly outperforms the scribble supervised competitors and partially surpasses the fully supervised ones.
- The scribble labels are labeled in V-D-T SOD dataset for the first time, which provide convenience for researches.

II. Related Works

A. Unified Framework Involved Salient Object Detection

Salient object detection task focuses on identifying prominent objects from their surroundings. When scenes are complex, the depth and thermal infrared cue from corresponding sensors can provide the supplement to RGB modality. Accordingly, RGB-D, RGB-T, and V-D-T SOD are derived. For example, DCF [32] calibrates unreliable raw depth maps first and then fuses both RGB and depth modalities. DSU [33] is an unsupervised RGB-D method. The pseudo labels are optimized via the depth cue. Both methods exploit the roles of depth images on RGB-D SOD task. Recently, some generalist models are proposed based on the consideration that RGB, RGB-D, and RGB-T SOD possess some commonalities. AiOSOD [34] is a unified model for RGB, RGB-D, and RGB-T SOD tasks. RGB-RGB, RGB-D, and RGB-T paired images are concatenated input in batch dimension to guarantee the consistency. Similarity, VSCode [35] simultaneously trains multi-modal salient object detection and camouflaged object detection tasks. Some 2D prompts are learnt to perceive the peculiarities on different modalities and tasks. UniSOD [36] individually trains the modal-aware prompts after pre-train stage for single-modal and multi-modal SOD tasks. Considering some binary segmentation tasks share vital fundamental similarities, EVP [20] presents a unified framework for salient object detection, forgery detection, defocus blur detection, shadow detection, and camouflage object detection. The task-specific knowledge is individually learnt by using a few of adapters [37]. An arbitrary modality SOD method AMSOD [38] is proposed for the robustness with different types and numbers of input modalities. Besides, some unsupervised methods are also proposed to achieve a unified framework, such as A2S-v2 [39], A2S-v3 [40].

Unlike the methods mentioned above, which mainly concentrate on fully supervised methods and unsupervised ones, our *SSFam* is the first unified model for scribble supervised methods among RGB, RGB-D, RGB-T, V-D-T SOD tasks. At first, modality is ranged from unimodal, bimodal to trimodal. Secondly, the scribbles are regarded as the prompt to assist segmentation. The designed modal-aware modulators and the siamese decoder are used to address the two challenges.

B. Weakly Supervised Salient Object Detection

The weakly supervised salient object detection methods use category, bounding boxes, points, subitizing, hybrid labels, scribbles to supervise the training of the models, which show the potential of leveraging the performance and annotation cost.

The category supervised methods [41], [42], [43], [44], [45] optimize the class activation maps (CAM) which supervise the training of saliency model. MSW [46] and JSM [45] further integrate the caption information besides image-level tags. In JSM [45], RGB images are used to predict the saliency maps which are further refined by estimated depth information. Next, initial handcrafted pseudo labels and the refined saliency maps are weighted combined by category predictions to form the updated pseudo labels. Last, a saliency network is trained by the updated pseudo labels. The box supervised methods [47], [48] update pseudo labels guided by bounding boxes. The point supervised method [49] integrates the edge information to generate pseudo labels due to extremely weak point signals. The subitizing based method [50] introduces counting information into SOD. The method via hybrid labels [51] utilizes a large number of coarse labels generated by the traditional unsupervised method and a small number of real labels.

The scribble based methods use sparse labels indicating foreground and background to achieve the whole object segmentation. WSSA [10], DENet [52], and RGBTScribble [53] propose scribble annotation datasets for RGB, RGB-D, and RGB-T SOD tasks, respectively. The methods explore the structure integrity under the guidance of edges and superpixels. Based on WSSA [10], SCWSSOD [11] design some loss functions to optimize the propagation of sparse annotations and ensure the consistency among different scales.

In our work, we use scribble annotation to train the whole model and test the image without scribble. Due to the strong feature discrimination ability of SAM, our method achieves impressive performance by combining the scribble loss [53] with our proposed consistency loss which transfers the knowledge with prompt to the ones with no prompt.

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C. Improvements Involved Segment Anything Model

SAM has demonstrated its prowess across various fields in terms of unimodal input [15], [18], [54], [55]. To achieve bimodal application, ViPT [56] and ProTrack [57] adapt the pre-trained RGB-based foundation model to multi-modality tracking tasks by learning visual prompt tokens. DUALPATH [58] transfers the parameters of the image foundation model to video domain by adapters. In audio-visual understanding tasks, a trainable latent audio-visual hybrid adapter LAVISH [59] is proposed to adapt unimodal ViTs to bimodal domain.

Facing our application with combinations of different modalities involved unimodal, bimodal, and trimodal, LoRAs with low rank properties are adopted as modal-aware modulators. They are attached to SAM to attain modal-specific knowledge and meet different input demands.

III. Method

A. SSSOD Task Description

To reduce the cost of pixel-level annotations, scribble supervised salient object detection (SSSOD) task takes as the supervision signals the scribbles which are sparse labels indicating foreground and background for training. Suppose SSSOD dataset $D = \{(\mathbf{x}_i, \mathbf{y}_i^{\mathrm{s}})\}_{i=1}^{N}$, where $\mathbf{x}_i$ is input image involved unimodal RGB, bimodal RGB-D and RGB-T, and trimodal V-D-T, $\mathbf{y}_i^{\mathrm{s}}$ is scribble annotation with 1 representing foreground, 2 denoting background, and 0 indicating unknown pixels, respectively, and N is the number of training samples. Then, the SSSOD is to train the model $\mathcal{F}$ with the parameters Ω and minimize the loss between predicted saliency maps and scribbles according to the formula:

$$\mathcal{L}(D, \Omega) = \sum_{i=1}^{N} \mathcal{L}_w(\mathcal{F}_\Omega(\mathbf{x}_i), \mathbf{y}_i^{\mathrm{s}})) \tag{1}$$

where the scribble loss $\mathcal{L}_w$ follows [53], and it is defined as:

$$\mathcal{L}_w = \mathcal{L}_{pce} + \mathcal{L}_{lsc} + \mathcal{L}_{sl} \tag{2}$$

where the partial cross entropy loss ($\mathcal{L}_{pce}$) supervises predicted saliency pixels only corresponding to scribble annotations, the local saliency coherence loss ($\mathcal{L}_{lsc}$) enforces similar pixels in the adjacent region to share consistent saliency scores, and the smoothness loss ($\mathcal{L}_{sl}$) guarantees the local smoothness and salient distinction along image edges.

B. SSSOD Solution

As a segmentation task, SSSOD can be implemented by Segment Anything Model (SAM). However, due to the lack of domain knowledge, zero-shot performance is poor. Moreover, SAM is pretrained on RGB images instead of bimodal and trimodal images. It is challenging to adapt to input with combinations of different modalities. To solve the challenge, different modal-aware modulators are designed to attain modal-specific knowledge which cooperates with modal-agnostic information extracted from frozen SAM encoder for a better feature ensemble.

Besides, the training of SSSOD lies on sparse scribble annotation rather than pixel-level ones. For a better performance under the less supervision, the model need more prompts. Fortunately, SAM provides the point prompt interface to indicate the position of segmentation. Some points randomly extracted from scribbles can be regarded as the prompt input of SAM. However, there is no scribble in the testing stage. The gap between the training with prompt and the testing with no prompt becomes the second challenge. To solve the challenge, a siamese decoder is tailored to transfer the knowledge from the decoding branch with prompt to the decoding branch with no prompt.

Accordingly, we propose an SSSOD family for unimodal, bimodal, and trimodal images, named *SSFam*, by attaching modal-aware modulators to SAM encoder and transforming SAM decoder to a siamese decoder structure. Then, the parameters Ω of the model $\mathcal{F}$ is defined as:

$$\Omega = \{\theta_{\mathrm{E}}^{*}, \theta_{\mathrm{M}}, \theta_{\mathrm{D}}^{\mathrm{p}}, \theta_{\mathrm{D}}^{\mathrm{n}}\} \tag{3}$$

where θ_{E}^{*} is the parameters of frozen image and prompt encoders in SAM, θ_{M} is the parameters of modulators, $\theta_{\mathrm{D}}^{\mathrm{p}}$ is the parameters of the decoder with prompt, and $\theta_{\mathrm{D}}^{\mathrm{n}}$ is the ones of the decoder with no prompt.

C. Two Challenges in SSFam

Challenge 1 Adapting SAM to input with combinations of different modalities: In the real-world scenes, image, depth and thermal infrared sensors are combinational equipped. The three sensors are complementary for detecting salient objects. For example, the color image provides more texture information, depth image offers some spatial geometrical cues, and thermal infrared image supplies scene perception by the temperature differences especially under the dark, rainy, and foggy environments. As a result, RGB, RGB-D, RGB-T, and V-D-T SSSOD has attracted a lot of research interests. Facing the input challenge with combinations of different modalities, *SSFam* equips modal-aware modulators to attain modal-specific knowledge. Specifically, θ_{M} is defined as:

$$\theta_{\mathrm{M}} = \begin{cases} \{\theta_{\mathrm{M}}^{\mathrm{v}}\}, & unimodal \\ \{\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{d}/}\theta_{\mathrm{M}}^{\mathrm{t}}\}, & bimodal \\ \{\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{d}}, \theta_{\mathrm{M}}^{\mathrm{t}}\}, & trimodal \end{cases} \tag{4}$$

where superscript indicates the categories of modality, v, d, t denotes RGB, depth, and thermal infrared modality, respectively.

Take as an example RGB-D SSSOD, the model structure of the encoding in *SSFam* is shown in the encoder part of Fig. 2. A shared and frozen SAM image encoder is used to extract modal-agnostic features, and two groups of modal-aware modulators are used to learn modal-specific features. The modal-specific feature and the modal-agnostic feature are aggregated by element-wise scaled sum. At last, the features from two modalities are summed up.

The modal-aware modulators can be the Adapters [37], LoRAs [60], SSF [61] and the other fine-tuning solutions [62]. Since the number of the parameters in LoRAs are less, LoRAs are chosen as modulators. Specifically, the input images $\mathbf{x}_i$ are fed into the frozen encoder E of SAM. The encoder is

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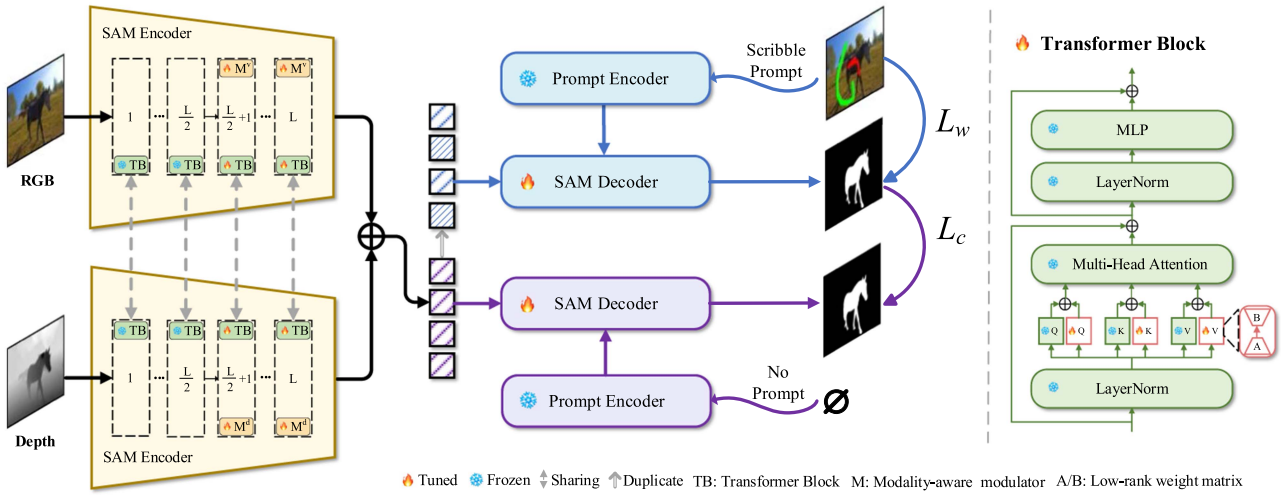


Fig. 2. The proposed scribble supervised salient object detection model. In the encoding part, a shared and frozen SAM encoder is used to extract modal-agnostic features, and some modal-aware modulators are designed to obtain modal-specific ones. Both features are aggregated by element-wise scaled sum in each block. Last, the features from two modalities are summed up. In the decoding part, a siamese decoder which consists of a decoder with prompts and a decoder with no prompt is proposed to transfer the parameters with prompts in the training to the ones with no prompt in the testing. When testing, the decoder with no prompt is only needed.

a ViT transformer structure with multiple blocks which consist of the multi-head self-attention layers and the feed forward neural layers. Each block $\mathrm{E}_l(l=1,\ldots,L)$ outputs the features $\mathbf{F}_l(l=1,\ldots,L)$. The output of previous block is fed into the next block. In the vanilla multi-head self-attention layer,

$$\mathbf{Q}=\mathbf{W}_{\mathrm{Q}}\mathbf{F}_l,\mathbf{K}=\mathbf{W}_{\mathrm{K}}\mathbf{F}_l,\mathbf{V}=\mathbf{W}_{\mathrm{V}}\mathbf{F}_l \tag{5}$$

$$\mathbf{H}=\mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{\mathrm{d}}}\right)\mathbf{V} \tag{6}$$

where $\mathbf{H}$ is the output of multi-head self-attention layer. It is will fed into feed forward neural layer to generate the output of the transformer block.

$$\mathbf{F}_{l+1}=\mathrm{FFN}(\mathbf{H}) \tag{7}$$

where $\mathrm{FFN}(\cdot)$ denotes the feed forward neural layer.

The designed modal-aware modulator introduces low-rank trainable parameters into the multi-head attention layer.

$$\mathbf{Q}^m=\mathbf{W}_{\mathrm{Q}}^m\mathbf{F}_l,\mathbf{K}^m=\mathbf{W}_{\mathrm{K}}^m\mathbf{F}_l,\mathbf{V}^m=\mathbf{W}_{\mathrm{V}}^m\mathbf{F}_l \tag{8}$$

where $m\in\{\mathrm{v},\mathrm{d}\}$ represents modalities in RGB-D SSSOD, and $\mathbf{W}_n^m(n\in\{\mathrm{Q},\mathrm{K},\mathrm{V}\})$ is composed of the original frozen parameters $\mathbf{W}_n$ and the learnable modal-aware parameters.

$$\begin{cases}\mathbf{W}_{\mathrm{Q}}^m=\mathbf{W}_{\mathrm{Q}}+\mathbf{W}_{\mathrm{Qa}}^m\mathbf{W}_{\mathrm{Qb}}^{m\ \mathrm{T}}\\ \mathbf{W}_{\mathrm{K}}^m=\mathbf{W}_{\mathrm{K}}+\mathbf{W}_{\mathrm{Ka}}^m\mathbf{W}_{\mathrm{Kb}}^{m\ \mathrm{T}}\\ \mathbf{W}_{\mathrm{V}}^m=\mathbf{W}_{\mathrm{V}}+\mathbf{W}_{\mathrm{Va}}^m\mathbf{W}_{\mathrm{Vb}}^{m\ \mathrm{T}}\end{cases} \tag{9}$$

where $\mathbf{W}_{na}^m\in\mathbb{R}^{d\times r}$ and $\mathbf{W}_{nb}^m\in\mathbb{R}^{d\times r}$, $r\ll d$. It is achieved by a down-projection linear layer followed by an up-projection linear layer.

At last the features in the last layers of all the modalities are aggregated by addition operation.

$$\mathbf{F}=\sum_{m\in\{v,d\}}\mathbf{F}_L^m \tag{10}$$

Note that it is easily inferred from above discussion about RGB-D example to the others. $m\in\{\mathrm{v}\}$ in RGB, $m\in\{\mathrm{v},\mathrm{t}\}$ in RGB-T, and $m\in\{\mathrm{v},\mathrm{d},\mathrm{t}\}$ in V-D-T SSSOD.

Challenge 2 Eliminating the gap between training with prompt and testing with no prompt: SAM supports point prompt for the better segmentation. The randomly extracted points from scribble labels are fed into SAM in the training stage. However, there is no prompt in the testing. To eliminate the gap, a siamese decoder is proposed to transfer the parameters with prompts in the training to the ones with no prompt in the testing. As shown in the decoder part of Fig. 2, the feature $\mathbf{F}$ from the encoder is duplicated and fed into prompt decoder branch and no prompt decoder branch, respectively. Then, our *SSFam* is learnt under the loss:

$$\begin{aligned}\mathcal{L}(D,\Omega)=&\sum_{i=1}^{N}\mathcal{L}_w(\mathcal{F}_{\Omega^{\mathrm{p}}}(\mathbf{x}_i,\mathbf{y}_i^{\mathrm{s}}),\mathbf{y}_i^{\mathrm{s}})\\&+\mathcal{L}_c(\mathcal{F}_{\Omega^{\mathrm{n}}}(\mathbf{x}_i),\mathcal{F}_{\Omega^{\mathrm{p}}}(\mathbf{x}_i,\mathbf{y}_i^{\mathrm{s}}))\end{aligned} \tag{11}$$

In formula 11, the first loss is scribble supervision loss, which is computed between predicted saliency maps generated from the prompt branch with the parameters $\Omega^{\mathrm{p}}=\{\theta_{\mathrm{E}}^*,\theta_{\mathrm{M}},\theta_{\mathrm{D}}^{\mathrm{p}}\}$ and scribble labels $\mathbf{y}_i^{\mathrm{s}}$. Note that the input of the prompt branch include the input image $\mathbf{x}_i$ and scribble prompt $\mathbf{y}_i^{\mathrm{s}}$. The second loss is a consistency loss $\mathcal{L}_c$ which is computed between predicted saliency maps from the no prompt branch with the parameters $\Omega^{\mathrm{n}}=\{\theta_{\mathrm{E}}^*,\theta_{\mathrm{M}},\theta_{\mathrm{D}}^{\mathrm{n}}\}$ and the ones from the prompt branch with the parameters Ω^p. The consistency loss transfers the knowledge with prompt to the ones with no prompt, and it is defined

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Algorithm 1: SSFam Training Process.

Input: The training set $D = \{(\mathbf{x}_i, \mathbf{y}_i^{\mathrm{s}})\}_{i=1}^{N}$, $\mathbf{x}_i$ is RGB, RGB-D, RGB-T, or V-D-T image and $\mathbf{y}_i^{\mathrm{s}}$ is scribble label;
Output: Model $\mathcal{F}$ with the parameters Ω;

Load SAM pretrained parameters, including image encoder and prompt encoder θ_{E}^{*}, decoder $\theta_{\mathrm{D}}^{\mathrm{p}}$, and another copied decoder $\theta_{\mathrm{D}}^{\mathrm{n}}$;
Initialize modal-specific modulators $\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{d}}, \theta_{\mathrm{M}}^{\mathrm{t}}$ randomly;
if $\mathbf{x}_i$ is RGB **then**
  $\theta_{\mathrm{M}} = \theta_{\mathrm{M}}^{\mathrm{v}}$;
else if $\mathbf{x}_i$ is RGB-D **then**
  $\theta_{\mathrm{M}} = \{\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{d}}\}$;
else if $\mathbf{x}_i$ is RGB-T **then**
  $\theta_{\mathrm{M}} = \{\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{t}}\}$;
else
  $\theta_{\mathrm{M}} = \{\theta_{\mathrm{M}}^{\mathrm{v}}, \theta_{\mathrm{M}}^{\mathrm{d}}, \theta_{\mathrm{M}}^{\mathrm{t}}\}$;
end if
$\Omega = \{\theta_{\mathrm{E}}^{*}, \theta_{\mathrm{D}}^{\mathrm{p}}, \theta_{\mathrm{D}}^{\mathrm{n}}, \theta_{\mathrm{M}}\}$;//all the parameters
$\Omega^{\mathrm{p}} = \{\theta_{\mathrm{E}}^{*}, \theta_{\mathrm{D}}^{\mathrm{p}}, \theta_{\mathrm{M}}\}$; //prompt branch parameters
$\Omega^{\mathrm{n}} = \{\theta_{\mathrm{E}}^{*}, \theta_{\mathrm{D}}^{\mathrm{n}}, \theta_{\mathrm{M}}\}$;//no prompt branch parameters
$\Omega^{\mathrm{pt}} = \Omega^{\mathrm{p}} - \theta_{\mathrm{E}}^{*}$;//trainable prompt branch parameters
$\Omega^{\mathrm{nt}} = \Omega^{\mathrm{n}} - \theta_{\mathrm{E}}^{*}$;//trainable no prompt branch parameters
repeat
Take gradient descent step on
$\bigtriangledown_{\Omega^{\mathrm{pt}}} \sum_{i=1}^{N} (\mathcal{L}_w(\mathcal{F}_{\Omega^{\mathrm{p}}}(\mathbf{x}_i, \mathbf{y}_i^{\mathrm{s}}), \mathbf{y}_i^{\mathrm{s}}))$
$+ \bigtriangledown_{\Omega^{\mathrm{nt}}} \sum_{i=1}^{N} (\mathcal{L}_c(\mathcal{F}_{\Omega^{\mathrm{n}}}(\mathbf{x}_i), F_{\Omega^{\mathrm{p}}}(\mathbf{x}_i, \mathbf{y}_i^{\mathrm{s}})))$
until converged
return $\mathcal{F}_{\Omega}$.

as:

$$\mathcal{L}_c = \mathcal{L}_{wbce} + \mathcal{L}_{iou} \tag{12}$$

where $\mathcal{L}_{wbce}$ is the weighted binary cross entropy loss and $\mathcal{L}_{iou}$ is weighted IoU loss [63].

In the end, the training process of *SSFam* is presented in Algorithm 1. When testing, the decoder with no prompt is only needed.

IV. Experiments

A. Datasets and Evaluation Metrics

To verify the effectiveness of *SSFam*, we conduct four groups of comparison experiments involved RGB, RGB-D, RGB-T, and V-D-T scribble supervised SOD. A summary of datasets is illustrated in Table I.

In the unimodal **RGB** SSSOD comparison experiments, the training dataset adopts 10,553 samples from S-DUTS [10]. The testing datasets include ECSSD [64] dataset (1,000 samples), DUTS-Test [41] dataset (5,019 samples), HKU-IS [65] dataset (4,447 samples), DUT-OMRON [66] dataset (5,168 samples), and PASCAL-S [67] dataset (850 samples).

In the bimodal **RGB-D** SSSOD comparison experiments, the training dataset adopts 2,185 paired RGB and depth images from NJU2K-S and NLPR-S [52]. The testing datasets include NLPR [68] with 300 images, NJU2K [69] with 500 images, STERE [70] with 1,000 images, SIP [71] with 929 images, DES [72] with 135 images, LFSD [73] with 100 images, and SSD [74] with 80 images.

TABLE I
Summary of Training and Testing Datasets

Input Type	Dataset Name	#Train	#Test
RGB	S-DUTS [10]	10,553	/
	ECSSD [64]	/	1,000
	DUTS-Test [41]	/	5,019
	HKU-IS [65]	/	4,447
	DUT-OMRON [66]	/	5,168
	PASCAL-S [67]	/	850
RGB-D	NJU2K-S [52]	1,485	/
	NLPR-S [52]	700	/
	NLPR-Test [68]	/	300
	NJU2K-Test [69]	/	500
	STERE [70]	/	1,000
	SIP [71]	/	929
	DES [72]	/	135
	LFSD [73]	/	100
	SSD [74]	/	80
RGB-T	RGBT-S [53]	2,500	/
	VT821 [75]	/	821
	VT1000 [76]	/	1,000
	VT5000-Test [77]	/	2,500
V-D-T	VDT-2048-S(*Our proposed*)	1,048	/
	VDT-2048-Test [30]	/	1,000

In the bimodal **RGB-T** SSSOD comparison experiments, the training dataset adopts 2,500 paired RGB and thermal infrared images from RGBT-S [53]. The testing datasets include VT821 [75] with 821 images, VT1000 [76] with 1,000 images, and VT5000 [77] with 2,500 images.

In the trimodal **V-D-T** SSSOD comparison experiments, we relabel VDT-2048 SOD [30] training set via scribbles, which includes 1,048 paired V-D-T images, namely VDT-S dataset. The remaining 1,000 samples of VDT-2048 are used to be tested.

Four evaluation metrics are used, including S-measure (S) [78], max F-measure (F_β) [79], max E-measure (E_ξ) [80], and mean absolute error (M) [81].

B. Implementation Details

The code is implemented in Pytorch toolbox on a PC with an NVIDIA RTX 3090 GPU. SAM uses ViT-L version, so L is 24. The modal-aware modulation is only conducted in the last half layers. During training, each image is uniformly resized to 1024 $\times$ 1024. No data augmentation strategy is adopted. The batch size is 1 and the initial learning rate is 5e-5. The model converges around 20 epochs by Adam optimizer [82]. The training time is about 10 hours.

C. Comparison With State-of-The-Art Methods

1) Quantitative Comparisons: The comparison results in RGB, RGB-D, RGB-T, and V-D-T SSSOD are shown in Tables II, III, IV, and V, respectively. In each table, we list some fully supervised methods, unsupervised methods, point supervised methods, box supervised methods, category supervised

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TABLE II
QUANTITATIVE COMPARISONS ON FIVE RGB DATASETS AND THEIR OVERALL PERFORMANCE

Methods	Source	Supervision	**Overall**				ECSSD				DUTS-Test			
			S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓
BIPG [86]	TMM21	Full	[.888]	[.874]	[.927]	[.038]	.929	.945	.960	.029	.896	.886	.937	.033
DPNet [87]	TIP22	Full	[.867]	[.837]	[.910]	[.046]	.918	.926	.951	.037	.869	.841	.914	.043
EDN [88]	TIP22	Full	[.902]	[.887]	[.938]	[.034]	.938	.948	.965	.027	.909	.898	.948	.030
BBRF [83]	TIP23	Full	[.900]	[.892]	[.941]	[.031]	.939	.957	.972	.022	.909	.906	.952	.025
MENet [89]	CVPR23	Full	[.893]	[.876]	[.929]	[.034]	.928	.939	.956	.031	.905	.895	.944	.028
AiOSOD [34]	Arxiv24	Full	[.882]	[.859]	[.923]	[.044]	.928	.939	.962	.034	.882	.856	.927	.041
A2Sv3 [40]	IJCAI24	Un	[.856]	[.839]	[.911]	[.048]	.905	.926	.952	.038	.848	.827	.908	.047
PSOD [49]	AAAI22	Point	[.861]	[.825]	[.895]	[.048]	.914	.921	.924	.036	.853	.811	.887	.045
SBBs [47]	TIP21	Box	[.807]	[.756]	[.868]	[.068]	.851	.855	.894	.072	.789	.722	.851	.073
WSS [41]	CVPR17	Category	[.764]	[.696]	[.818]	[.099]	.811	.823	.869	.104	.748	.654	.795	.100
ASMO [42]	AAAI18	Category	[.731]	[.638]	[.780]	[.111]	.802	.797	.853	.110	.697	.614	.772	.116
MSW [46]	CVPR19	Category*	[.779]	[.707]	[.823]	[.097]	.827	.840	.884	.096	.759	.684	.814	.091
MFNet [43]	ICCV21	Category	[.787]	[.739]	[.851]	[.077]	.834	.854	.885	.084	.775	.710	.839	.076
NSAL [44]	TMM22	Category	[.792]	[.749]	[.855]	[.074]	.834	.856	.884	.077	.781	.730	.849	.073
WSSA [10]	CVPR20	Scribble	[.818]	[.778]	[.881]	[.061]	.865	.871	.920	.059	.804	.755	.873	.062
SCWSSOD [11]	AAAI21	Scribble	[.844]	[.827]	[.905]	[.051]	.882	.902	.932	.049	.841	.823	.907	.049
SCOD [90]	AAAI23	Scribble	[.856]	[.826]	[.907]	[.050]	.902	.915	.946	.040	.854	.822	.910	.047
SB [12]	TMM23	Scribble	[.854]	[.835]	[.902]	[.048]	.890	.906	.917	.046	.853	.834	.901	.045
Ours	TMM24	Scribble	**[.903]**	**[.893]**	**[.943]**	**[.033]**	**.942**	**.954**	**.972**	**.023**	**.904**	**.898**	**.946**	**.030**

Methods	Source	Supervision	HKU-IS				DUT-OMRON				PASCAL-S			
			S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓
BIPG [86]	TMM21	Full	.924	.936	.963	.025	.845	.797	.882	.051	.861	.859	.910	.060
DPNet [87]	TIP22	Full	.905	.911	.950	.034	.825	.754	.868	.056	.848	.833	.891	.071
EDN [88]	TIP22	Full	.934	.942	.969	.023	.865	.820	.899	.045	.877	.874	.916	.056
BBRF [83]	TIP23	Full	.932	.946	.972	.020	.861	.822	.899	.044	.878	.879	.927	.049
MENet [89]	CVPR23	Full	.927	.939	.965	.023	.850	.792	.879	.045	.872	.869	.915	.054
AiOSOD [34]	Arxiv24	Full	.918	.927	.961	.033	.845	.789	.882	.055	.863	.853	.910	.064
A2Sv3 [40]	IJCAI24	Un	.898	.913	.954	.033	.821	.771	.872	.062	.843	.843	.900	.067
PSOD [49]	AAAI22	Point	.902	.907	.952	.032	.824	.748	.854	.064	.853	.831	.857	.065
SBBs [47]	TIP21	Box	.854	.843	.920	.056	.776	.695	.835	.074	-	-	-	-
WSS [41]	CVPR17	Category	.822	.821	.896	.079	.725	.603	.768	.109	.744	.715	.791	.139
ASMO [42]	AAAI18	Category	-	-	-	-	.752	.622	.776	.101	.717	.693	.772	.149
MSW [46]	CVPR19	Category*	.818	.814	.895	.084	.756	.609	.763	.109	.768	.713	.790	.133
MFNet [43]	ICCV21	Category	.846	.851	.921	.059	.742	.646	.803	.087	.770	.751	.817	.115
NSAL [44]	TMM22	Category	.854	.864	.923	.051	.745	.648	.801	.088	.768	.756	.822	.110
WSSA [10]	CVPR20	Scribble	.865	.865	.931	.047	.785	.708	.841	.068	.797	.780	.861	.092
SCWSSOD [11]	AAAI21	Scribble	.882	.898	.943	.038	.812	.756	.868	.060	.820	.821	.881	.078
SCOD [90]	AAAI23	Scribble	.904	.909	.954	.032	.809	.739	.856	.068	.841	.835	.898	.069
SB [12]	TMM23	Scribble	.895	.906	.949	.034	.818	.762	.868	.060	.825	.827	.853	.073
Ours	TMM24	Scribble	**.934**	**.942**	**.970**	**.022**	**.873**	**.838**	**.915**	**.044**	**.876**	**.875**	**.921**	**.055**

In fully supervised methods, the best results are underlined. In scribble supervised methods, the best results are bold. Pink background indicates our overall performance. '-' indicates no evaluation metrics report. 'Category*' denotes category+caption supervision.

methods and some scribble supervised methods. Meanwhile, we also collect the overall performances on multiple testing datasets in the "Overall" column in terms of 5 datasets in RGB, 7 datasets in RGB-D, and 3 datasets in RGB-T. For a fair comparison, the saliency maps of other methods are provided by authors or generated by the released codes and we evaluate them with the same evaluation code.

From the experimental results shown in all the datasets and the overall performance, our model consistently achieves non-trivial performance among scribble supervised methods. In terms of the overall performance, our model improves the MAE evaluation metric about 0.02 in RGB and RGB-D dataset, and 0.01 in RGB-T dataset. As we are the first to achieve V-D-T scribble supervised method, no comparison method can be obtained. For the clearer demonstration, we present the overall performance comparisons in Fig. 3. From the figure, we can find our model significantly outperforms competitors.

Furthermore, our method is also very close to fully supervised methods and only lower than the best fully supervised method in MAE, e.g., RGB BBRF [83], RGBT PRLNet [84], V-D-T QSFNet [85] about 0.002, 0.001, and 0.001, respectively. It is worth mentioning that all metric values on overall performance of RGB-D SSSOD have surpassed those of fully supervised methods. The remarkable improvement benefits from the strong segmentation ability of fine-tuned SAM by modal-aware modulators and the knowledge transfer from prompt decoder to no prompt decoder.

Last, our method is compared with some other weakly supervised methods, such as box [47], point [49], category [41], [42], [43], [44], [45], [46], and some unsupervised unified methods [40]. Existing methods constructed on the convolution neural networks can't win over our method based on the large model SAM with transformer backbones and our proposed modulation including modality-aware modulators and the siamese decoder.

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TABLE III
QUANTITATIVE COMPARISONS ON SEVEN RGB-D DATASETS AND THEIR OVERALL PERFORMANCE

Methods	Source	Supervision	**Overall**				NLPR				NJU2K				STERE			
			S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓
JL-DCF [26]	TPAMI21	Full	[.905]	[.904]	[.949]	[.040]	.926	.917	.964	.023	.911	.913	.948	.040	.911	.907	.949	.039
SwinNet [91]	TCSVT22	Full	[.921]	[.925]	[.956]	[.031]	.941	.936	.974	.018	.935	.938	.963	.027	.919	.918	.956	.033
PCIR-Net [92]	ACM MM23	Full	[.915]	[.920]	[.952]	[.033]	.935	.931	.970	.019	.927	.931	.958	.029	.920	.920	.957	.031
HRTransNet [93]	TCSVT23	Full	[.920]	[.926]	[.956]	[.030]	.942	.936	.974	.016	.933	.939	.963	.026	.921	.919	.956	.030
HiDANet [94]	TIP23	Full	[.910]	[.924]	[.944]	[.034]	.930	.929	.961	.021	.926	.939	.954	.029	.911	.921	.946	.035
CATNet [25]	TMM23	Full	[.922]	[.927]	[.958]	[.029]	.940	.934	.972	.018	.932	.937	.960	.026	.921	.922	.958	.030
AiOSOD [34]	Arxiv24	Full	[.915]	[.915]	[.951]	[.036]	.927	.911	.958	.025	.925	.923	.957	.033	.915	.909	.950	.038
DFormer [95]	ICLR24	Full	[.925]	[.929]	[.959]	[.027]	.942	.936	.973	.016	.937	.943	.967	.023	.923	.920	.956	.030
A2Sv3 [40]	IJCAI24	Un	[.887]	[.889]	[.934]	[.042]	.905	.892	.949	.028	.881	.882	.924	.049	.892	.895	.941	.040
JSM [45]	NeurIPS21	Category*	[.745]	[.721]	[.820]	[.115]	.805	.771	.875	.060	.713	.711	.801	.133	.782	.772	.852	.096
DENet [52]	TIP22	Scribble	[.883]	[.854]	[.911]	[.047]	.913	.874	.936	.028	.899	.878	.922	.044	.887	.851	.914	.044
DHFR [96]	TIP23	Scribble	[.875]	[.891]	[.926]	[.047]	.904	.901	.950	.027	.893	.901	.936	.040	.884	.896	.935	.043
RPPS [97]	TPAMI24	Scribble	[.891]	[.890]	[.938]	[.053]	.911	.896	.953	.033	.893	.899	.940	.054	.882	.883	.931	.061
MIRV [98]	TCSVT24	Scribble	[.889]	[.887]	[.936]	[.043]	.914	.904	.957	.025	.890	.890	.934	.046	.891	.888	.939	.042
Ours	TMM24	Scribble	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	**.932**	**.965**	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	Source	Supervision	SIP				DES				LFSD				SSD			
			S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓
JL-DCF [26]	TPAMI21	Full	.892	.900	.949	.046	.936	.929	.975	.021	.863	.862	.900	.071	.860	.833	.902	.053
SwinNet [91]	TCSVT22	Full	.911	.927	.950	.035	.945	.945	.979	.016	.886	.889	.921	.059	.892	.878	.925	.040
PCIR-Net [92]	ACM MM23	Full	.899	.915	.939	.040	.942	.942	.977	.016	.888	.894	.923	.052	.874	.862	.923	.047
HRTransNet [93]	TCSVT23	Full	.909	.929	.949	.035	.947	.952	.982	.014	.877	.872	.912	.058	.898	.890	.934	.035
HiDANet [94]	TIP23	Full	.892	.919	.927	.043	.946	.952	.980	.013	-	-	-	-	.871	.854	.915	.047
CATNet [25]	TMM23	Full	.911	.928	.952	.034	.942	.940	.977	.017	.894	.892	.928	.051	-	-	-	-
AiOSOD [34]	Arxiv24	Full	.907	.922	.949	.038	.946	.940	.980	.018	.886	.886	.916	.061	.885	.866	.931	.044
DFormer [95]	ICLR24	Full	.915	.930	.953	.032	.948	.952	.984	.013	-	-	-	-	.894	.885	.934	.035
A2Sv3 [40]	IJCAI24	Un	.881	.889	.932	.044	.904	.902	.939	.025	.863	.874	.909	.065	.859	.845	.916	.050
JSM [45]	NeurIPS21	Category*	.697	.645	.774	.146	.820	.807	.875	.058	.767	.781	.833	.127	.652	.617	.751	.157
DENet [52]	TIP22	Scribble	.865	.841	.900	.056	.916	.887	.939	.029	.832	.814	.860	.085	.850	.799	.873	.061
DHFR [96]	TIP23	Scribble	.843	.874	.902	.064	.915	.930	.956	.022	.855	.878	.904	.066	.858	.869	.911	.051
RPPS [97]	TPAMI24	Scribble	.892	.892	.939	.050	.919	.914	.962	.027	.868	.877	.913	.078	.868	.856	.919	.056
MIRV [98]	TCSVT24	Scribble	.876	.879	.928	.049	.928	.921	.965	.019	.849	.844	.889	.072	.891	.873	.934	.042
Ours	TMM24	Scribble	**.927**	**.938**	**.958**	**.028**	**.930**	**.933**	**.967**	**.017**	**.891**	**.892**	**.920**	**.057**	**.902**	**.900**	**.945**	**.034**

In fully supervised methods, the best results are underlined. In scribble supervised methods, the best results are bold. Pink background indicates our overall performance. '**-**' indicates no evaluation metrics report. 'Category*' denotes category+caption supervision.

TABLE IV
QUANTITATIVE COMPARISONS ON THREE RGB-T DATASETS AND THEIR OVERALL PERFORMANCE

Methods	Source	Supervision	**Overall**				VT821				VT1000				VT5000			
			S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β ↑	E_ξ ↑	M↓	S↑	F_β↑	E_ξ ↑	M↓
TNet [29]	TMM22	Full	[.904]	[.894]	[.944]	[.030]	.899	.888	.938	.030	.929	.930	.966	.021	.895	.881	.937	.033
HRTransNet [93]	TCSVT23	Full	[.917]	[.909]	[.958]	[.023]	.906	.888	.941	.026	.938	.941	.975	.017	.912	.903	.956	.025
CAVER [99]	TIP23	Full	[.908]	[.894]	[.949]	[.025]	.898	.877	.934	.027	.938	.939	.973	.017	.899	.882	.944	.028
WaveNet [100]	TIP23	Full	[.919]	[.907]	[.955]	[.023]	.912	.895	.943	.024	.945	.945	.977	.015	.911	.896	.950	.026
PRLNet [84]	TIP23	Full	[.926]	[.923]	[.965]	[.022]	.917	.910	.954	.025	.944	.950	.980	.016	.921	.916	.962	.023
AiOSOD [34]	Arxiv24	Full	[.908]	[.892]	[.946]	[.030]	.905	.882	.937	.028	.941	.941	.974	.020	.895	.875	.938	.035
UniTR [101]	TMM24	Full	[.913]	[.914]	[.958]	[.021]	.901	.898	.941	.025	.938	.939	.975	.014	.907	.910	.956	.023
A2Sv3 [40]	IJCAI24	Un	[.886]	[.870]	[.932]	[.035]	.882	.866	.922	.041	.923	.921	.959	.023	.872	.851	.924	.038
WSVSOD [102]	CVPR21	Scribble	[.831]	[.788]	[.892]	[.050]	.822	.762	.875	.052	.886	.870	.935	.035	.812	.763	.880	.055
DENet [52]	TIP22	Scribble	[.850]	[.805]	[.896]	[.045]	.813	.732	.839	.054	.906	.891	.947	.028	.839	.794	.895	.048
RGBTScribble [53]	ICME23	Scribble	[.892]	[.877]	[.942]	[.029]	.895	.878	.942	.027	.925	.922	.964	.020	.877	.859	.933	.033
Ours	TMM24	Scribble	**[.922]**	**[.914]**	**[.955]**	**[.023]**	**.919**	**.896**	**.943**	**.023**	**.943**	**.942**	**.973**	**.015**	**.914**	**.909**	**.951**	**.026**

In fully supervised methods, the best results are underlined. In scribble supervised methods, the best results are bold. Pink background indicates our overall performance.

TABLE V
QUANTITATIVE COMPARISONS ON VDT-2048 DATASET

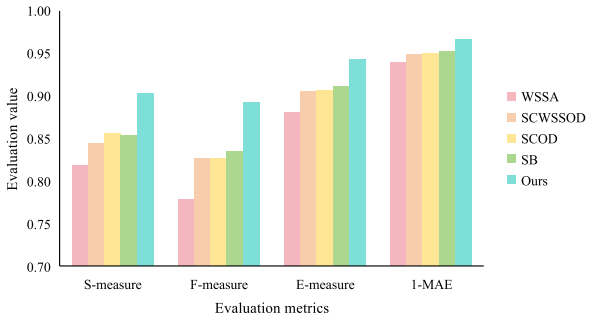
Methods	Source	Supervision	VDT-2048			
			S↑	F_β ↑	E_ξ ↑	M↓
HWSI [30]	TMECH23	Full	.932	.897	.981	.003
TMNet [103]	PR24	Full	.933	.910	.989	.003
FFANet [104]	PR24	Full	.935	.893	.989	.003
MFFNet [105]	TMM23	Full	.936	.914	.990	.002
QSFNet [85]	TIP24	Full	.943	.925	.993	.002
A2Sv3 [40]	IJCAI24	Un	.716	.545	.734	.006
Ours	TMM24	Scribble	.916	.884	.978	.003

In fully supervised methods, the best results are underlined. Our scribble supervised method is trained using our relabeled VDT-S SOD datasets.

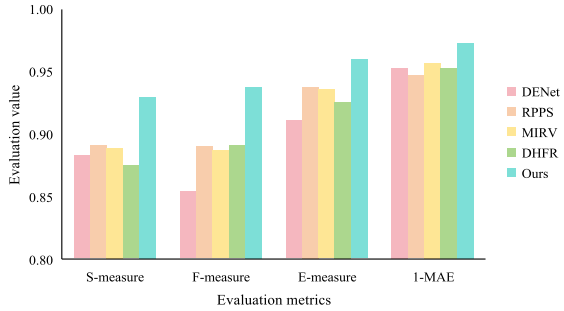
2) Visual Comparisons: Fig. 4 shows some visual examples for reflecting object, fine-grained object, multiple objects, and small objects in RGB SOD task. Fig. 5 shows some visual examples for small objects, multiple objects, hollow objects, and light interference in RGB-D SOD task. Fig. 6 shows some visual examples for fine-grained objects, low-quality RGB images, and small objects in RGB-T SOD task. Our results are obviously better, which indicate the advantage of our model. However, there are some failure cases. In the first two rows of Fig. 7, when foreground and background are similar and their depths are also close, some non-salient objects are detected. In the last two rows of Fig. 7, the fine-grained object parts, e.g., the legs of insects and the propeller, are missing. It can be inferred from the performances of the other scribble supervised methods that it is a great challenge to segment the complete object from sparse scribble labels.

3) Complexity Comparisons: Take RGB-D SSSOD as an example, Table VI presents complexity comparison. From the result, it can be observed that the computation cost of our model

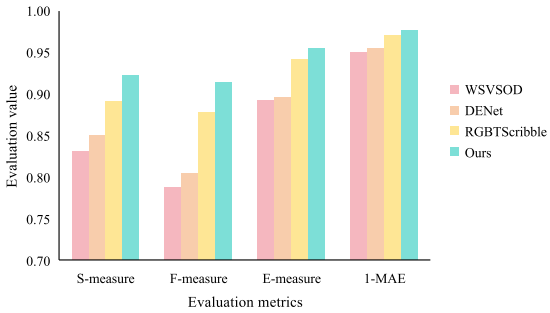
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(a) RGB scribble supervised SODs



(b) RGB-D scribble supervised SODs



(c) RGB-T scribble supervised SODs

Fig. 3. Comparisons of overall performance in RGB, RGB-D, and RGB-T scribble supervised salient object detection methods. No V-D-T SSSOD methods are compared because we are the first to introduce the scribble supervised method in V-D-T SOD.

TABLE VI
COMPLEXITY COMPARISONS WITH RGB-D SSSOD COMPETITORS IN TERMS OF INPUT RESOLUTION, COMPUTATION COST, AND MODEL SIZE

	DENet [52]	RPPS [97]	DHFR [96]	Ours
Input Resolution	352×352	256×256	224×224	1024×1024
FLOPs (G)	95.27	23.10	-	2635.70
Trainable Params (M)	37.14	53.39	54.5	9.24
Overall Params (M)	37.14	53.39	54.5	303.24

'-' indicates no evaluation metrics report.

is larger, because the core backbone SAM is a large model. However, we also find the trainable parameters are small because modal-aware modulators use the LoRAs with low-rank properties and the siamese decoder is two lightweight SAM decoders.

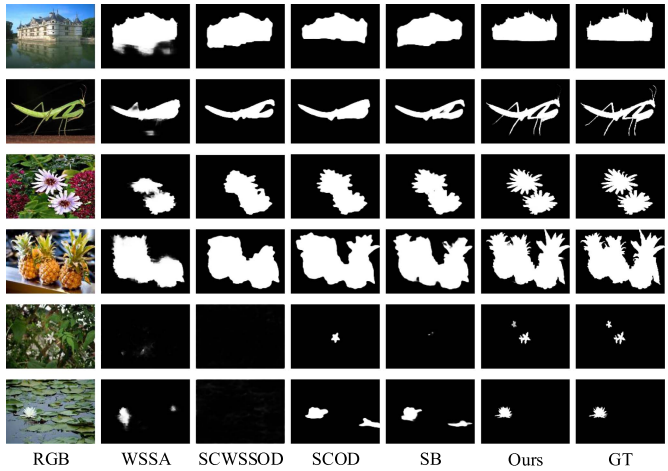


Fig. 4. Visual comparisons with RGB SSSOD competitors in some challenging cases: reflecting object (1st row), fine-grained object (2nd row), multiple objects (3rd-4th rows), and small objects (5th-6th rows).

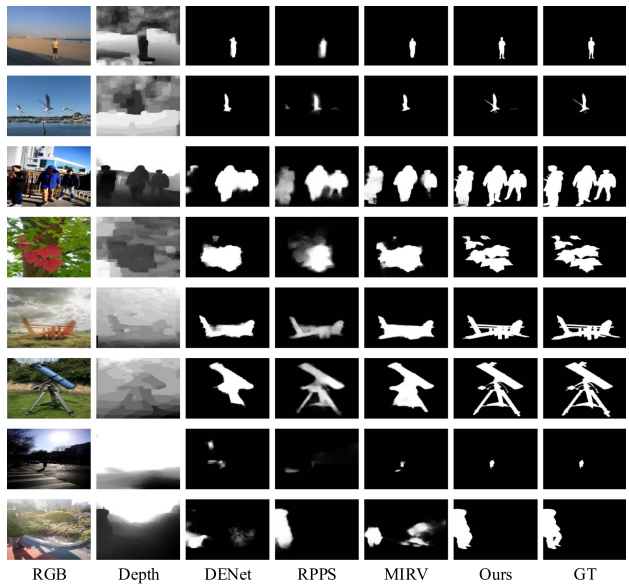


Fig. 5. Visual comparisons with RGB-D SSSOD competitors in some challenging cases: small objects (1st-2nd rows), multiple objects (3rd-4th rows), hollow objects (5th-6th rows), and light interference (7th-8th rows).

D. Ablation Studies

All the ablation studies are taken on RGB-D SSSOD task.

1) Ablation on Different Modulators: As SAM is a large model with a large number of parameters, the cost of full-tuning model is high, which is not achieved by common single-card GPU. Among the prevalent fine-tuning techniques, adapter and LoRA are compared as the modulators for RGB-D SSSOD task. From the results in "Overall" column of Table VII, we can see that fine-tuning is effective regardless adapter and LoRA, and LoRA is superior to adapter in the performance. Moreover, we list the number of trainable parameters when no modulators, using adapters, and using LoRAs. The number of parameters in two decoders is 7.74M. It adds the number of parameters in RGB and

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TABLE VII
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT THE MODULATORS

Methods	Trainable Params(M)	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
No modulator	7.74	[.912]	[.920]	[.949]	[.034]	.919	.920	.961	.023	.919	.921	.949	.034	.914	.921	.953	.034
Adapter modulators	13.81	[.919]	[.923]	[.953]	[.031]	.930	.926	.963	.019	.929	.931	**.957**	**.028**	.928	.936	.962	.027
LoRA modulators (Ours)	9.24	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	**.932**	**.965**	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	FLOPs (T)	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
No modulator	2.63	.904	.921	.942	.038	**.945**	**.944**	**.981**	**.013**	.879	.883	.913	.063	.884	.884	.938	.041
Adapter modulators	2.65	.905	.911	.943	.038	.929	.927	.969	.018	.884	.886	.912	**.057**	.886	.884	.931	.039
LoRA modulators (Ours)	2.64	**.927**	**.938**	**.958**	**.028**	.930	.933	.967	.017	**.891**	**.892**	**.920**	**.057**	**.902**	**.900**	**.945**	**.034**

The best results are bold, and pink background indicates our overall performance.

TABLE VIII
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT THE DECODER

Methods	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
scribble prompt train& no prompt test	[.915]	[.916]	[.949]	[.035]	.928	.916	.958	.022	.923	.927	.954	.033	.929	.931	.959	.027
no prompt train& no prompt test	[.920]	[.928]	[.955]	[.031]	.932	.925	.964	.020	.927	.937	**.957**	.031	.925	.942	**.966**	.029
siamese decoder (Ours)	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	**.932**	**.965**	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
scribble prompt train& no prompt test	.896	.901	.934	.047	.921	.917	.965	.022	.877	.877	.908	.064	.892	.880	.943	.039
no prompt train& no prompt test	.911	.916	.944	.036	**.931**	.929	**.968**	.018	.885	.887	.917	.060	.892	.891	.934	.038
siamese decoder (Ours)	**.927**	**.938**	**.958**	**.028**	.930	**.933**	.967	**.017**	**.891**	**.892**	**.920**	**.057**	**.902**	**.900**	**.945**	**.034**

The best results are bold, and pink background indicates our overall performance.

TABLE IX
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT THE MODALITIES

Methods	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
RGB	[.920]	[.927]	[.951]	[.031]	.934	.929	.964	.019	.928	.936	.954	.029	.931	.937	.960	.026
RGB-D (Ours)	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	**.932**	**.965**	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
RGB	.904	.918	.939	.038	.926	.928	.965	.019	.882	.885	.913	.060	.889	.882	.935	.040
RGB-D (Ours)	**.927**	**.938**	**.958**	**.028**	**.930**	**.933**	**.967**	**.017**	**.891**	**.892**	**.920**	**.057**	**.902**	**.900**	**.945**	**.034**

The best results are bold, and pink background indicates our overall performance.

TABLE X
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT THE NUMBER OF PROMPT POINTS EXTRACTED FROM SCRIBBLES

Methods	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
30 prompt	[.917]	[.921]	[.954]	[.032]	.928	.925	**.967**	.020	.924	.934	**.958**	.030	.918	.924	.958	.032
20 prompt	[.921]	[.928]	[.954]	[.029]	.935	**.932**	.966	.019	.927	.936	.955	**.028**	.929	.936	.959	**.025**
10 prompt (Ours)	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	**.932**	.965	**.018**	**.932**	**.941**	.957	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
30 prompt	.913	.916	.948	.034	.929	.929	**.970**	**.017**	.877	.885	**.920**	.062	.887	.872	.931	.042
20 prompt	.911	.922	.947	.034	.923	.928	.968	.019	.882	.891	.916	.060	.899	.897	**.946**	.036
10 prompt (Ours)	**.927**	**.938**	**.958**	**.028**	**.930**	**.933**	.967	**.017**	**.891**	**.892**	**.920**	**.057**	**.902**	**.900**	.945	**.034**

The best results are bold, and pink background indicates our overall performance.

depth adapters or LoRAs and the sums are 13.81M and 9.24M, respectively. It manifests the effectiveness of LoRAs compared to adapters.

2) Ablation on Decoder: To verify the effectiveness of our siamese decoder, we conduct the ablation on the decoder. Since SAM provide the prompt interface, scribble can be regarded as both supervision signals and prompts. However, there is no prompt in the testing stage. Therefore, the model which is trained with scribble prompt and tested with no prompt in the first row of Table VIII are obviously inferior to the model which is trained and tested both with no prompt in the second row. It reveals that there is a gap between training with prompts and testing with no

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TABLE XI
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT THE POSITION OF MODULATORS

Methods	Trainable Params(M)	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
L/3	8.74	[.920]	[.930]	[.956]	[.031]	.933	.929	.966	.020	.928	.934	.956	.030	.928	.944	.967	.028
2L/3	9.74	[.924]	[.930]	[.956]	[.029]	**.937**	**.932**	**.969**	**.018**	.931	.936	**.957**	**.028**	**.935**	.945	**.968**	**.024**
L/2 (Ours)	9.24	**[.930]**	**[.938]**	**[.960]**	**[.027]**	.936	**.932**	.965	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	.966	.025

Methods	FLOPs (G)	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
L/3	2633.10	.909	.922	.944	.037	.928	**.933**	.965	.018	.884	.891	**.922**	.062	.888	.885	**.946**	.042
2L/3	2638.10	.909	.917	.943	.037	.928	.926	**.968**	.018	.890	**.894**	.917	**.056**	**.902**	**.900**	.944	**.033**
L/2 (Ours)	2635.70	**.927**	**.938**	**.958**	**.028**	**.930**	**.933**	.967	**.017**	**.891**	.892	.920	.057	**.902**	**.900**	.945	.034

The best results are bold, and pink background indicates our overall performance.

TABLE XII
ABLATION STUDY ON RGB-D SOD DATASETS ABOUT INPUT RESOLUTION

Methods	Trainable Params(M)	Overall S↑	Overall F_β ↑	Overall E_ξ ↑	Overall M↓	NLPR S↑	NLPR F_β ↑	NLPR E_ξ ↑	NLPR M↓	NJU2K S↑	NJU2K F_β↑	NJU2K E_ξ ↑	NJU2K M↓	STERE S↑	STERE F_β↑	STERE E_ξ ↑	STERE M↓
256× 256	9.24	[.922]	[.928]	[.957]	[.029]	.935	**.933**	**.967**	**.018**	.925	.929	.955	.030	.919	.927	.957	.031
1024×1024 (Ours)	9.24	**[.930]**	**[.938]**	**[.960]**	**[.027]**	**.936**	.932	.965	**.018**	**.932**	**.941**	**.957**	**.028**	**.935**	**.947**	**.966**	**.025**

Methods	FLOPs (G)	SIP S↑	SIP F_β ↑	SIP E_ξ ↑	SIP M↓	DES S↑	DES F_β ↑	DES E_ξ ↑	DES M↓	LFSD S↑	LFSD F_β↑	LFSD E_ξ ↑	LFSD M↓	SSD S↑	SSD F_β↑	SSD E_ξ ↑	SSD M↓
256×256	245.5	.922	.933	.957	.029	**.935**	**.939**	**.971**	**.016**	**.892**	**.900**	**.927**	**.053**	.897	.871	.937	.037
1024×1024 (Ours)	2635.7	**.927**	**.938**	**.958**	**.028**	.930	.933	.967	.017	.891	.892	.920	.057	**.902**	**.900**	**.945**	**.034**

The best results are bold, and pink background indicates our overall performance.

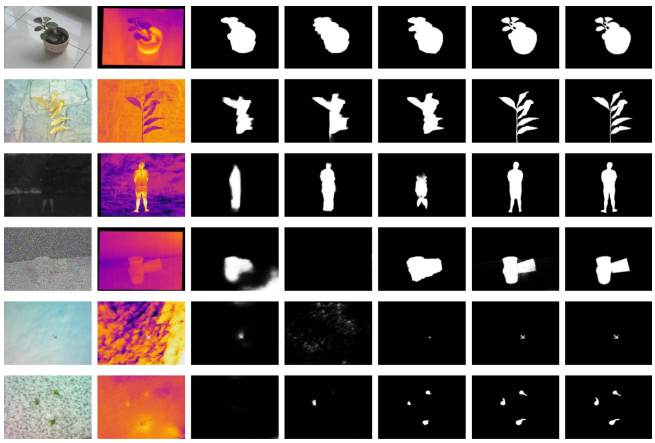


Fig. 6. Visual comparisons with RGB-T SSSOD competitors in some challenging cases: fine-grained objects (1st-2nd rows), low-quality RGB images (3rd-4th rows), and small objects (5th-6th rows).

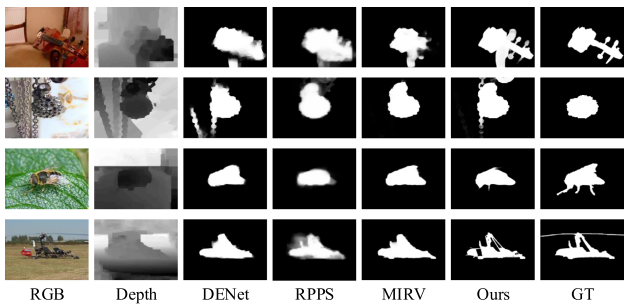


Fig. 7. Failure cases: severe background interference (1st-2nd rows), fine-grained details (3rd-4th rows).

prompt. Furthermore, the majority of the results in the second row of Table VIII are worse than ours in the third row, which indicating that our siamese decoder can transfer the knowledge of the trained model with prompts to the testing model with no prompt and deliver the better performance in the testing stage.

3) Ablation on Different Modalities: RGB-D SOD task is inspired by the fact that depth can provide the spatial cue when RGB image is not enough to provide effective information in some challenging scenes, for example, foreground and background are similar, and scenes are cluttered. The existing methods combine RGB and depth features to remedy the defect of unimodal input. To show the usefulness of depth cue in our model, we compare the performance with RGB modality and that with RGB and depth modalities. From the results in Table IX, we find depth modality plays an auxiliary role because the performance in MAE evaluation metric is slightly improved about 0.004 in overall. Moreover, RGB's performance has surpassed a few of existing fully supervised RGB-D methods in Table III. The above results consistently give the hint that the features extracted from SAM is very good so that the role of depth information is minor.

4) Ablation on the Number of Point Prompts From Scribbles: Table X gives the ablation study of the number of point prompts from scribbles. We find the model with 10 point prompts from scribbles is better than those with 20 and 30 points. Therefore, we randomly extract 10 points from scribbles as the prompt of SAM.

5) Ablation on the Position of Modal-Aware Modulators: Table XI gives the ablation study of the position of modal-aware modulators. L/3, 2L/3, and 2/L indicate that the modal-aware modulators are conducted in the last third, two-thirds, and half of the layer, respectively. As the number of layers increases so do

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the trainable parameters and FLOPs. However, the overall performances are not incremental. It can be inferred that the features in the latter half of layers of SAM transformer encoder are more important when adapting SAM to salient object detection task.

6) Ablation on the Input Resolution: Table XII gives the ablation study of the input resolution. The batchsize can be set as 16 when model adopts 256×256 input resolution in the training stage. The model has advantage in a few datasets, such as NLPR, DES, and LFSD. However, the overall performance is worse than that of the model with 1024×1024 input resolution. Meanwhile, the model with 256×256 input resolution is more friendly than that with 1024×1024 in computation cost. FLOPs of the former is 10 times less than that of the latter. Therefore, the model with 256×256 input resolution is also an alternative in limited computation resources.

V. Conclusion

In the work, *SSFam* is proposed based on the large model SAM for the scribble supervised salient object detection task in unimodal, bimodal, and trimodal images. The modal-aware modulators are responsible for attaining modal-specific information, adapting to the input with combinations of different modalities. The siamese decoder transfers the knowledge from prompt branch to no prompt branch, reducing the gap between the prompt training and no prompt testing. Experiments delivers the outstanding performance across RGB, RGB-D, RGB-T, and V-D-T SOD tasks. Due to the high computation cost of *SSFam* based on SAM, the real-time deployment of model is challenging, which is reserved for future research.

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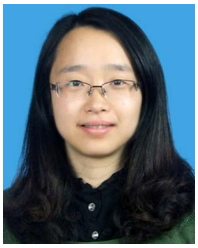
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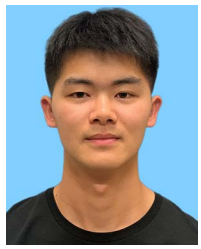
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